# COS 736 Big Data Engineering

Question 2C: Inference & Insights

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### Question 2C.1 – Top 3 Countries by Number of Purchases

The three countries with the highest number of purchase records in the dataset are shown below. The United Kingdom accounts for the large majority of all transactions.

| Rank | Country | Number of Purchases |
|---|---|---|
| 1 | United Kingdom | 495,478 |
| 2 | Germany | 9,495 |
| 3 | France | 8,557 |

### Question 2C.2 – Total Sales for the Top 3 Countries

Total sales were calculated as the sum of Quantity multiplied by UnitPrice for each of the three countries identified above.

| Rank | Country | Total Sales (£) |
|---|---|---|
| 1 | United Kingdom | £8,187,806.36 |
| 2 | Germany | £221,698.21 |
| 3 | France | £197,403.90 |

### Question 2C.3 – Most Sold Stock Item per Region

For each region, the stock item with the highest total quantity sold across all invoices was identified by grouping the data by Region and Description, then selecting the maximum.

| Region | Most Sold Stock Item | Total Qty Sold |
|---|---|---|
| Americas | RETRO COFFEE MUGS ASSORTED | 504 |
| Asia | RABBIT NIGHT LIGHT | 3,401 |
| Europe | WORLD WAR 2 GLIDERS ASSTD DESIGNS | 52,742 |
| Middle East | WOODLAND CHARLOTTE BAG | 130 |
| Oceania | MINI PAINT SET VINTAGE | 2,916 |
| Other | RAIN PONCHO | 407 |